ZF-DDOS: An Enhanced Statistical-Based DDoS Detection Approach using Integrated Z-Score and Fast-Entropy Measures

Mahmoud Hassan
Departemnt of Computer Engineering and Artifitial Intelligence
Military Technical Colleague
Cairo, Egypt
mahmoudhassan82068@gmail.com

Khaked Metwally
Departemnt of Computer Engineering and Artifitial Intelligence
Military Technical Colleague
Cairo, Egypt
k.metwally@mtc.edu.eg

Mohamed A. Elshafey
Departemnt of Computer Engineering and Artifitial Intelligence
Military Technical Colleague
Cairo, Egypt
m.shafey@mtc.edu.eg

2024 6th International Conference on Computing and Informatics (ICCI) | 979-8-3503-7387-5/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICCI61671.2024.10485097

***Abstract*— The fast emerging of Internet technologies and 5G connectivity played an imperative role in our daily lives. This allowed to incredible increase in the Internet usage, so, a large amount of information is exchanged between various connected devices daily. However, this development introduced a great cyber security challenge for all individuals and enterprises. Evolution of new and different types of network attacks grows quickly to compromise one or more of the Confidentiality, Integrity, and Availability (CIA) Triad principles that guarantee the security of information systems. Distributed Denial of Service (DDoS) attack is one of the most destructive attacks that targets the availability in the CIA Triad. Many research works have been proposed targeting DDoS attacks detection in recent networks. However, the detection accuracy still a challenging task. This paper presents an enhanced statistical-based DDoS detection approach that mainly relies on the integration of Z-Score and Fast-Entropy statistical measurements. Moreover, the proposed approach integrates additional statistical features that contribute significantly on higher DDoS detection accuracy over statistical based benchmarks. In experiments, the proposed approach has been tested on CIC-DDOS-2019 dataset and resulted markable improvements in DDOS detection accuracy of 25% and 10% over Fast-Entropy- and Z-Score statistical-based benchmarks, respectively.**

***Keywords*— CIA Triad, Distributed Denial of Service (DDoS), Z-Score, Fast-Entropy, CIC-DDOS-2019.**

I. INTRODUCTION

The proliferation of the connected devices, over the internet, increased the number of attacks on users that impact their information exchange and disclose their sensitive data. Recent cyber threats are becoming more sophisticated and more severe than ever before, exploiting the existence of security vulnerabilities in the inter-connected systems.

Distributed Denial of Service (DDoS) attacks poses an immense threats to the connected systems and/or devices over the Internet. DDoS is a type of cyber-attack, in which malicious actors aimed to render a computer or a device unavailable to its intended users via interrupting the device's normal functioning. DDoS attacks typically function by overwhelming or flooding a targeted machine with too many requests until normal traffic becomes unable to be processed, resulting in a Denial-of-Service to additional users [1]. The distributed nature in the aforementioned attack stemmed from disrupting the normal functioning of a website or network by overwhelming it with traffic from multiple sources such as a botnet network [2].

According to the findings of NETSCOUT Threat Intelligence Report, in the first half of 2023, NETSCOUT observed a staggering total of ~7.9 million DDoS attacks, representing a 31% increase compared to 1H 2022 and a 15.6% increase compared to 2H 2022. This represents an unbelievable 44000 DDoS attacks per day [3]. Moreover, aaccording to the findings of secure list by Kaspersky in Q3-2022, the top four countries in terms of resources attacked remained unchanged from the previous reporting period. The US (39.60%) remained in the first place. Mainland China's share (13.98%) increased by almost the same amount, up 6.31%. Germany (5.07%) remains in the third and France (4.81%) in the fourth place [4]. It is obvious that the attackers use botnets to perform DDoS attacks against other countries to impact their businesses return.

Over the last decades, Intrusion Detection Systems (IDSs) have been evolved to secure the network traffic against malicious users/attacks. Many research work have been proposed IDSs to detect and mitigate DDoS attacks in network. IDSs can be classified into two detection techniques; misuse-based techniques and anomaly-based techniques. In misuse-based detection technique, the performed detection is mainly relied on predefined patterns or signatures of previously known attacks to identify and block malicious traffic, also, it is known as signature-based technique. The major problem for signature-based detection is its ineffectiveness against zero-day attacks or new attack variations that have not been previously identified and included in the signature database [5]. On the other side, anomaly-based detection focuses on identifying deviations from normal or expected network behavior. It establishes a baseline of normal behavior and then flags any traffic or events that fall outside this baseline as potential anomalies. However, the prediction and prevention of DDoS attacks are still a challenging issue [6]. Recent researchers have developed many IDSs applying different approaches and techniques, e.g., statistical-based techniques, Machine Learning-based (ML), and Deep Learning-based (DL) techniques [7].

In this paper, a hybrid approach using statistical-based analysis techniques is presented to protect the network against DDoS attacks more effectively. Although statistical methods are very primitive methods, however, it is considered as a light weight technique in feature preprocessing and feature

Authorized licensed use limited to: UNIVERSIDADE DE BRASILIA. Downloaded on October 22,2026 at 00:29:27 UTC from IEEE Xplore. Restrictions apply.

extraction compared to ML/DL techniques [8]. Furthermore, it can be combined with other techniques to construct a more powerful hybrid detection approach.

In the context of IDS, statistical-based techniques apply various methods to measure the collected network traffic central tendency as well as the dispersion of the data, e.g., mean, median, mode, standard deviation (std), quantiles, boxplot, etc. Statistical measurements are characterized by its fast response, low processing speed and computation overheads, moreover, statistical measures give insights on the collected network traffic and its dynamicity changes. In ML-based techniques, various models work effectively on a small amount of dataset and determine the statistical properties of attacks before classifying or scoring them [9]. However, they required periodical parameters update to reflect the conducted changes in the attack patterns. Also, DL-based techniques can effectively identify DDoS attacks but it requires recent large labelled datasets, that reflect current and new DDoS attacks patterns, to build precise and accurate detection model [10].

It was clear from the literature survey that the main challenges in DDoS attacks detection can be summarized as follows: outdated datasets, long-time detection, offline analysis of the adopted dataset, the low detection accuracy and existing detection models complexity. The contribution in this paper as follow:

1- To design and develop an enhanced light-weight statistical hybrid approach for DDoS detection that integrates Z-Score and Fast-Entropy statistical based measures.

2- To evaluate and validate the proposed approach against statistical based benchmarks in the literature survey. A performance analysis is presented in terms of various evaluation metrics, e.g., accuracy, precision, recall, and F-score and compared against benchmark techniques.

The rest of the paper is organized as follows: Section II provides the related research works, while section III presents the proposed DDoS detection approach. In section IV, the experimental work environment, the datasets, and the comparative results are discussed. Finally, section V illustrates the conclusion and future work.

II. Related Work

In cybersecurity, various DDoS attack detection techniques have been proposed including, misuse-based detection, anomaly-based detection and various hybrid techniques have been proposed as well. Statistical-based analysis and Machine/Deep learning-based methods can be classified as anomaly-based detection [11]. In this section recent research in the context of statistical-based analysis and ML/DL based methods will be discussed.

In [12], the authors proposed DDoS detection model using Fast-Entropy trained on the CAIDA dataset. The proposed solution consist of three main objectives which are: flow aggregation for doing flow-based attack detection, Fast-Entropy calculations for each connection and adaptive threshold algorithm to improve detection accuracy. Based on the obtained results, the authors stated that during flooding attack, the flow count will increase, flow entropy drops drastically for attack case and it is stable for non-attack cases. However, Fast-Entropy calculations may be sensitive to parameter choices such as the time window for flow count calculations, choosing the appropriate parameters, for the detection process, is crucial. In addition, this method primarily is designed for flow count-based anomaly detection and it may not capture certain types of DDoS attacks that manifest differently. In addition, the experimental results are conducted on an earlier dataset CAIDA-2007 [13].

In [14], Hussein Majed et al. proposed DDoS detection model that mainly relied on the Z-Score and co-variance measures to detect DDoS traffic as a deviation from normal traffic. The proposed model required collect net flow using software installed on router or external device. the collected NetFlow statistics are aggregated per destination IP then apply statistical data analysis based on the Z-Score to allow normal traffic or activate mitigation on malicious traffic. However, the adopted threshold needs to be updated manually, each time a bandwidth upgrade takes place Which might lead to high false positives and false negatives rates. In addition, this method was not experimented with a well-known dataset.

In [15], the authors presented an entropy-based method that can identify both high and low-rate DDoS attacks on a Software Defined Network (SDN) controller, independent of the quantity of targets or attackers. The proposed method extended the Rényi joint entropy to analyze network traffic flow and identify different rates of DDoS attack traffic flow. Their experimental results outperformed the Shannon entropy metrics-based on the Early Detection of DDoS Attacks in Software Defined Networks Controller (EDDSC) approach in terms of detection rate by utilizing two packet header features and generalized Rényi joint entropy.

An effective method in [16] was proposed for detecting DoS and DDoS attacks based on the Exponential Smoothing (ES) scheme and the Continuous Ranked Probability Score (CRPS) statistical measure. The authors illustrated that the difference between the distribution of typical traffic and a new observation was measured using the CRPS. The ES technique was used to detect anomalies in CRPS readings since it is sensitive enough to pick up minor changes. Additionally, the CRPS-ES approach used a non-parametric decision threshold to appropriately detect abnormalities, which was obtained by kernel density estimation. The proposed approach was tested and evaluated based on DARPA99, MAWI, and ICMPv6 datasets[17,18,19] Comparing the integrated CRPS-ES against other widely used algorithms, the achieved results suggested that it was performed well.

The authors in [20] presented entropy-based methods to identify and mitigate DDoS attacks. Several traits were extracted from the payload and used to pinpoint the attack. The proposed method demonstrated in three steps: active, preparatory, and nominal mitigation. While the typical pattern baseline was obtained at the nominal phase in an attack-free environment, the assault samples were recognized during the preliminary stage. The countermeasure against the attack vector was then put into practice at the active mitigation step.

The authors in [21] introduced the Hurst coefficient, autoregression coefficient, and coefficient of variation as three one-parameter based statistical techniques for DDoS attack detection. They evaluated the coefficients using both real attack records and simulated data. Moreover, the authors suggested the so-called predicted σ-tunnel for early machine recognition of the attack. However, the obtained results need applying more prediction techniques in order to enhance the early detection of attacks.

Authorized licensed use limited to: UNIVERSIDADE DE BRASILIA. Downloaded on October 22,2026 at 00:29:27 UTC from IEEE Xplore. Restrictions apply.

The paper in [22] adopted network-based statistical signal processing techniques to present a straight forward but efficient DDoS assault detection strategy. The properties of the packets in the network traffic were utilized to create statistical distributions, and the difference between these models was used to detect attacks. Network traffic is split up into time periods of equal length. The empirical probability distribution of the network traffic was defined as the empirical distribution of transport protocol types and transport protocol flags of network packets within time frames. The k-means algorithm was used to construct distributions from each training data window and cluster them. The attack indicator was defined as the lowest divergence between the empirical distributions created from the test data and the clustered distributions.

In order to detect DDoS assaults based on TCP protocol and UDP protocol, the paper in [23], originally employed 102 features and 49 features for DDoS attacks based on TCP and UDP, respectively. The detection accuracy and running time of their proposed GBDT method were compared with those of K-Nearest Neighbors (KNN), Naive Bayes (NB), Support Vector Machine (SVM), and (Multi-Layer Perceptron) MLP. According to the obtained results, the GBDT algorithm achieved a greater detection accuracy than the other algorithms and required less testing time. Using the GBDT algorithm as the evaluation criterion, the random forest and Pearson wrap (RFPW) feature selection algorithm employed a search approach that combined random forest scores with Pearson correlation coefficients. In this work, RFPW was compared with conventional feature selection and dimensionality reduction algorithms.

Recent advancements and current issues in IDs technology are examined in [24] and an adaptive ensemble learning methodology is suggested. The authors built a multi-tree algorithm that modified the percentage of training data and created numerous decision trees. They selected multiple base classifiers, e.g., decision tree, Random Forest (RF), KNN, and Deep Neural Network (DNN), and created an ensemble adaptive voting algorithm to enhance the overall detection effect. The authors validated their method with the NSL-KDD Test; the Multi-tree algorithm yields an accuracy of 84.2%, and the adaptive voting strategy yields a final accuracy of 85.2%.

In [25], the authors proposed an IDs that can identify DDoS attacks using a proposed deep learning model and preprocessing techniques. A number of DL models based on Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNN), and DNNs have been assessed using CIC-DDoS-2019 for real-time and detection performance.

The authors provided and examined in [26] the effectiveness of 25 time-based features for binary and multiclass classification in the detection and classification of 12 different forms of DDoS attacks. Additionally, they conducted tests comparing the effectiveness of one deep learning classifier and eight conventional ML classifiers under two distinct circumstances. Their results demonstrated that most of the tested models produced results of about 70% accuracy in categorizing specific DDoS attack types and about 99% accuracy in detecting DDoS attacks on both the time-based and control tests.

From the above mentioned papers, it seems that author used machine statistical, machine learning and deep learning methods to detect DDoS attack.

III. The Proposed ZF-DDoS Hybrid Detection Approach

Statistical-based analysis considered to be one of the most powerful methods for DDoS detection [27]. It uses thresholds to differentiate between normal and abnormal traffic. Using multiple (hybrid) metrics and thresholds in combination helps to improve the reliability and accuracy of the system [28].

The proposed approach in this research mainly relies on two important statistical measurements; Z-Score that represents a data point but in a normalized Gaussian distribution, i.e., it measures exactly how many standard deviations above or below the mean a data point is and Fast-Entropy which related with the measure of disorder or uncertainty. These scores may represent outliers or non-outliers. They point out but, in a more convenient way, where the measurements lie. The Z-Score is calculated by:

$$z = (x-\mu)/\sigma \,, \tag{1}$$

where x is the observed value, μ is the mean of the sample, and σ is the standard deviation of the sample. It is calculated for each attribute, e.g., for TCP_Flows.

Entropy is a scientific concept, as well as a measurable physical property, that is most commonly associated with a state of randomness of data. Entropy-based approaches have significant benefits in DDoS detection [29]. When the monitored network runs in normal way, the entropy values are relatively smooth, otherwise, the entropy value of one or more features would change significantly. Entropy is calculated by the following formula:

$$H(X) = -\sum_{i=1}^{n} P(x_i) \log_2 P(x_i) \,, \tag{2}$$

where $P(x_i)$is probability of each outcome x_i .

Thus, both Fast-Entropy and Z-Score can play valuable and benefitable complementary roles in such an accurate detection model. Combining the advantages of both of them in a hybrid model can lead to a higher detection accuracy over benchmarks. Implementing Z-Score in the proposed hybrid approach provides:

- Ensured versatility characteristic; Z-Score can be applied to various attributes beyond flow counts, such as packet sizes or inter-arrival times, providing versatility in anomaly detection.
- Easy deployment, scalability, robustness, and the flexibility.
- Efficient detection well known and unknown ("Zero-day") attacks with low false alarm rate.

While integrating Fast-Entropy supports:

- More effective and sensitive in detecting anomalies due to the sensitive nature the Fast-Entropy in the distribution of flow counts.
- Ability for real time continuously monitoring network traffic, as Fast-Entropy can be calculated efficiently and in real-time

The detection method adopted in the proposed approach is based on a threshold based comparison with the calculated statistical measures. The thresholds for both Fast-Entropy and

Authorized licensed use limited to: UNIVERSIDADE DE BRASILIA. Downloaded on October 22,2026 at 00:29:27 UTC from IEEE Xplore. Restrictions apply.

Z-Score that indicate abnormal behaviors are established and set based on historical data or statistical analysis. Anomalous conditions will be triggered when either Fast-Entropy or Z-Score surpasses its respective threshold. This research proposed a ZF-DDOS hybrid approach that is fast and reliable solution based on the NetFlow statistics on the CIC-DDOS-2019 dataset that ensures the detection of DDOS attacks with higher accuracy by combining statistical measures e.g., Fast-Entropy, Z-Score, co-variance and other statistical methods.

Figure 1 illustrates the block diagram of the proposed ZF-DDOS hybrid detection approach.

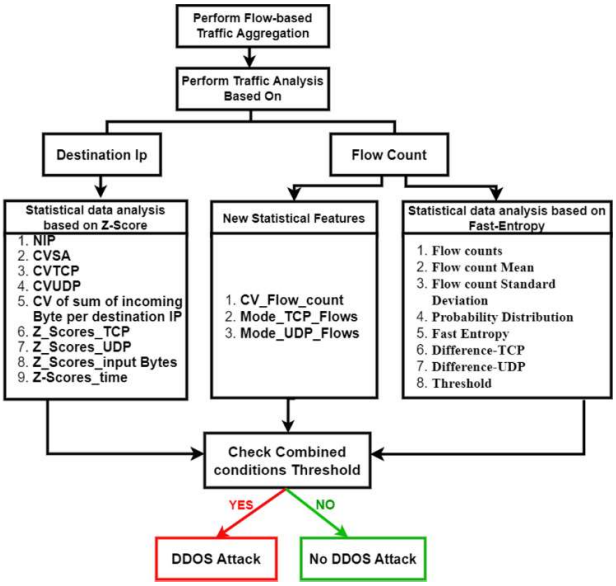


Fig.1. The block diagram of the proposed ZF-DDOS hybrid approach.

As shown in Figure 1, the approach is mainly relied on flow-based analysis and follows three statistical analysis based stages that execute concurrently as follows:

1) Statistical analysis based on Z-Score

In this stage, the proposed approach depends on the measures of Z-Score and Co-Variance (CV) to detect DDoS traffic. The detection is measured as a deviation from normal traffic based on the aggregated flow statistics. The Z-Score measures exactly how many standard deviations above or below the mean value, while the CV is the ratio of the standard deviation to the mean.

The Z-Score is a well-known method for the detection of outliers. Applying the Z-Score on each value in the aggregated flow will clearly reveal the outlier, but each type of an attack has its own characteristics. Firstly, the following features are calculated for each destination address of the dataset:

- *The sum of received bytes*,
- *The sum of received packets*,
- *The count of source addresses that communicated with each destination address*,
- *The sum of time duration*,
- *The count of UDP Flows*,
- *The count of TCP Flows*,
- *The count of ICMP Flows*.

Then, the CV metrics of the corresponding features are calculated that is considered the initial step to detect normal traffic statistics and build the base-line model values:

- *CV(SA) - the coefficient variation of the number of source addresses per destination IP.*
- *CV(TCP) - the coefficient variation of sequence of TCP flows.*
- *CV(UDP) - the coefficient variation of sequence of UDP flows .*
- *CV(ICMP) - the coefficient variation of sequence of ICMP flows.*

Finally, Z-Score metric can be calculated on the previous features for each destination IP address (aggregated flow). A DDoS attack is detected if the obtained Z-Score values of these features are greater or equal to the corresponding threshold [14].

In the proposed approach, the detection rule of Z-Score (ZS) follows the conditions:

IF ZS(SA) > CV(SA) then
If ZS(TCP) < CV(TCP) and ZS_Time_Duration>0 then TCP_DDoS_Attack_Detected_on_IP.
else if ZS(UDP)<CV(UDP) and ZS (Input Byte) <CV (Input Byte) then UDP_DDoS Attack_Detected_on_IP. (3)
else if ZS(ICMP) > CV(ICMP) and ZS (Input Byte) < CV (Input Byte) then ICMP_DDoS_Attack_Detected_on_IP.

2) Statistical data analysis based on Fast-Entropy

In this stage, the flow count is calculated for each connection at a particular time interval. Then, the Fast-Entropy statistical measure will be calculated for flow count of each connection. From the observation it is clear that the Fast-Entropy value is considerably reduced for particular connection and time interval of which flow count is large value compared to rest. DDoS attack is detected, when the difference between entropy of flow count at each instant (D(i,t)) and mean value of entropy in that time interval is greater than the threshold value (ß). Since the threshold value is updated adaptively based on traffic pattern condition, the accuracy of detection has been improved [12]. The detection rule of Fast-Entropy follows:

$$DDoS_Attack_Detected\ (i,t) < ß * \sigma \quad (4)$$

The Fast-Entropy threshold value is updated adaptively according to the Fast-Entropy value based on traffic pattern condition following the conditions:

If H(i,t) > 1.5μ then increase ß by 1
else if 0.5μ < = H (i,t) < = 1.5μ maintain current ß , (5)
else if H (i,t) < 0.5μt decrease ß by1.

where H(i,t) is Fast-Entropy, μ : mean of flow count during a particular time.

3) Statistical analysis based on new additional features

In this stage, new statistical features are calculated and added to the propsoed detection appraoch in order to improve the detection accuracy. The newly calculated features are :

a) *CV_Flow_Counts*: coefficient of variation of flow count suggests variability in the number of flows over time. When calculate the CV for flow counts in a network, a higher CV indicates that the flow

Authorized licensed use limited to: UNIVERSIDADE DE BRASILIA. Downloaded on October 22,2026 at 00:29:27 UTC from IEEE Xplore. Restrictions apply.

counts are more spread out from the mean. A lower CV would indicate more stability or consistency in flow count.

b) *Mode_TCP_flows*: The mode is the value that appears most frequently in a dataset, finding that mode can help to identify the most common number of TCP flows.

c) *Mode_UDP_flows*: The same as Mode_TCP_flows but for UDP flows.

The ZF-DDOS Detection Rule follows the conditions:

If ZS(SA) > Mode_TCP_Flow then TCP DDoS Attack Detected on IP.
else if ZS(SA) >Mode_UDP_Flow then UDP DDoS Attack Detected on IP. (6)

It is worth mentioning that, the rational behind selecting the mode was the calculated mean, or average, is sensitive to extreme values (outliers) and can be skewed by them.

The flowchart of proposed ZF-DDOS hybrid approach, as a three parallel stages, is shown in Figure 2.

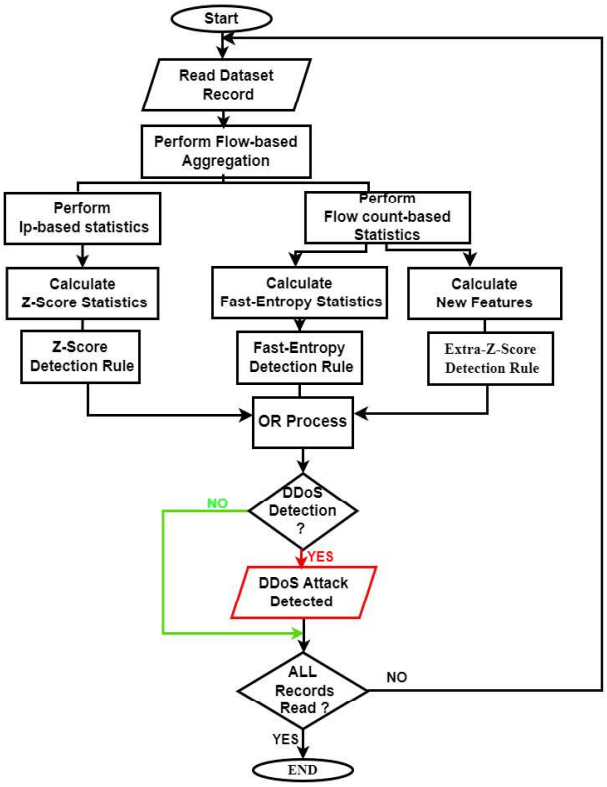


Fig. 2. The flowchart of the proposed ZF-DDOS hybrid approach.

After performing the statistical analysis on the aggregated flow-based features in the three parallel stages, a threshold detection phase will be applied in each stage separately, as shown in Figure 2. The final decision of the proposed ZF-DDoS hybrid approach will be issued using an ORed voting operation among the three conducted statistical results as follows:

$$D = Z \vee F \vee N, \quad (7)$$

where Z, F and N represent the decisions of Z-Score, Fast-Entropy, new proposed features, separately, and D is the final decision.

IV. Experimental Work

In this section, we will illustrate the adopted dataset, the working environment, performance metrics, the conducted experiments results, and comparative analysis on the results.

A. Dataset Description

In the proposed ZF-DDOS hybrid approach, the evaluation process adopted CIC-DDOS-2019 dataset which is contributed by Canadian Institute for Cybersecurity (CIC). CIC-DDoS-2019 contains the most updated DDoS attacks and benign traffic as well. This dataset contains 12 different types of DDoS attacks (e.g. NTP, DNS, LDAP, MSSQL, NetBIOS, SNMP, SSDP, UDP, UDP-Lag, WebDDOS, SYN and TFTP, PortScan, NetBIOS, LDAP, MSSQL, UDP, UDP-Lag and SYN) in the format of PCAP files and CSV files executed. This dataset was captured during two days one for training and the other for testing [30].

B. Enviromental Work

The experimental work has been carried out on PC with intel-core i7 processor, 32 GByte memory, that runs on windows11-64bit operating system. The proposed approach was developed in Python 3.10.9 (64-bit) using anaconda environment framework and Jupiter notebook.

The conducted experiments have been carried out on three models (Z-Score, Fast-Entropy, ZF-DDOS) using the same CIC-DDOS-2019 dataset. As mentioned before, the adopted dataset organized in two days, the dataset has been shuffled between the two days and the three models during the training and the testing experiments.

C. Performance Evaluation Metrics

For performance evaluation. the confusion matrix has been calculated for binary classification: malware or not malware. The confusion matrix has four metrics:

- True Positive (TP): indicates how the system correctly predicted that a sample is positive (malware).
- True Negative (TN): indicates how the system correctly predicted that a sample is negative (not malware).
- False Positive (FP): indicates how the system incorrectly predicted that a sample is positive (malware), but it is actually negative.
- False Negative (FN): indicates how the system incorrectly predicted that a sample is negative (not malware), but it's actually positive.

Moreover, the confusion metrics are then used to calculate the *accuracy, Precision, Recall, and F1-score* metrics as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (8)$$

$$Precision = \frac{TP}{TP + FP} \quad (9)$$

$$Recall = \frac{TP}{TP + FN} \quad (10)$$

$$F1 - score = \frac{2 * Precision * \text{Recall}}{Precision + \text{Recall}} \quad (11)$$

Authorized licensed use limited to: UNIVERSIDADE DE BRASILIA. Downloaded on October 22,2026 at 00:29:27 UTC from IEEE Xplore. Restrictions apply.

Where:

- *Accuracy* metric measures the overall correctness of the model. It is the ratio of correctly predicted instances (both true positives and true negatives) to the total number of instances.
- *Precision* metric focuses on the accuracy of the positive predictions. It is the ratio of correctly predicted positive observations to the total predicted positives. Precision is particularly relevant when the cost of false positives is high.
- *Recall* metric measures the ability of the model to capture all the relevant instances. It is the ratio of correctly predicted positive observations to all actual positives. *Recall* is crucial when the cost of false negatives is high.
- *F1-score*: The F1 score is the harmonic mean of *precision* and *recall*. It provides a balance between *precision* and *recall*. The F1 score is useful when there is an uneven class distribution.

D. Results and Comparative Analysis

The proposed ZF-DDOS hybrid approach has been evaluated against two state-of-the-art statistical-based benchmarks [12] [14] . The first benchmark relies on the Z-Score and co-variance in DDoS detection while the other benchmark adopted Fast-Entropy. Moreover, in [12], the authors adopted a subset of CAIDA 2007 dataset for the tested dataset. While the authors in [14] implemented a customized dataset that was created using a virtual CentOS machine created with VMware ESXi 6.0. This machine has two network interfaces, the first one is for NetFlow collection, while the second is for traffic management.

The conducted experiments for performance evaluation of the proposed ZF-DDOS hybrid approach has been achieved using the unified CIC-DDOS-2019 dataset against the benchmark's models. Table I represents the Z-Score model results in a detailed fashion. As shown in the table, the *Precision, Recall* and *F1-Score* results are calculated for both TCP and UDP traffic. In addition, the Z-Score has the ability to detect TCP DDoS attack with high accuracy but not as in detecting of UDP traffic.

Table II represents the detailed results of the Fast-Entropy based benchmark. In Table II, the Fast-Entropy model evaluates the overall system traffic and achieves less accuracy than the Z-Score.

The results of the proposed ZF-DDOS hybrid approach in details are presented in Table III, in which the results of Precision, Recall and F1-Score are calculated for both TCP and UDP traffic as case of Z-Score model. Moreover, ZF-DDOS is superior in detecting them with 100% accuracy that exceeds the 83% accuracy of Z-Score and the 73% accuracy of Fast-Entropy.

Table IV shows the comparative results between the three competing approaches, while Figure 3 illustrates the enhancement ratios of the proposed ZF-DDOS hybrid approach over the benchmark approaches in terms of *Accuracy, Precision, Recall, and F1-Score*.

TABLE I. Z-SCORE DETAILED EVALUATION RESULTS

Day	File	Number of records	Accuracy	Precision		Recall		F1-Score	
				TCP	*UDP*	*TCP*	*UDP*	*TCP*	*UDP*
1st Day	DrDoS_DNS	4,320,541	83.45%	87.7%	0.00%	83.3%	0.00%	85.5%	0.00%
	DrDoS_LDAP	2,181,542	83.45%	87.7%	0.00%	83.3%	0.00%	85.5%	0.00%
	DrDoS_MSSQL	4,524,498	83.45%	87.7%	0.00%	83.3%	0.00%	85.5%	0.00%
	DrDoS_NetBIOS	4,094,986	83.45%	87.7%	0.00%	83.3%	0.00%	85.5%	0.00%
	DrDoS_NTP	1,217,007	83.45%	87.7%	0.00%	83.3%	0.00%	85.5%	0.00%
	DrDoS_SNMP	5,161,377	83.45%	87.7%	0.00%	83.3%	0.00%	85.5%	0.00%
	DrDoS_SSDP	2,611,374	83.45%	87.7%	0.00%	83.3%	0.00%	85.5%	0.00%
	DrDoS_UDP	3,136,802	83.45%	87.7%	0.00%	83.3%	0.00%	85.5%	0.00%
	Syn	1,582,681	83.45%	87.7%	0.00%	83.3%	0.00%	85.5%	0.00%
2nd Day	LDAP	2,113,234	83.45%	87.7%	0.00%	83.3%	0.00%	85.5%	0.00%
	MSSQL	5,775,786	83.45%	87.7%	0.00%	83.3%	0.00%	85.5%	0.00%
	NetBIOS	3,455,899	83.45%	87.7%	0.00%	83.3%	0.00%	85.5%	0.00%
	Portmap	191,694	83.45%	87.7%	0.00%	83.3%	0.00%	85.5%	0.00%
	Syn	4,320,541	83.45%	87.7%	0.00%	83.3%	0.00%	85.5%	0.00%
	UDP	3,782,206	83.45%	87.7%	0.00%	83.3%	0.00%	85.5%	0.00%
	UDPLag	725,165	83.45%	87.7%	0.00%	83.3%	0.00%	85.5%	0.00%
Mixed files	DrDoS_LDAP, UDP	2,906,707	83.45%	87.7%	0.00%	83.3%	0.00%	85.5%	0.00%
	Syn, UDP	5,364,887	83.45%	87.7%	0.00%	83.3%	0.00%	85.5%	0.00%
	DrDoS_MSSQL,MSSQL	10,300,284	83.45%	87.7%	0.00%	83.3%	0.00%	85.5%	0.00%
Average			83.45%	87.7%	0.00%	83.3%	0.00%	85.5%	0.00%

Authorized licensed use limited to: UNIVERSIDADE DE BRASILIA. Downloaded on October 22,2026 at 00:29:27 UTC from IEEE Xplore. Restrictions apply.

TABLE II. FAST-ENTROPY DETAILED EVALUATION RESULTS

Day	File	Number of records	Accuracy	Precision	Recall	F1-Score
1st Day	DrDoS_DNS	4,320,541	93.10%	89%	100%	94%
	DrDoS_LDAP	2,181,542	78.95%	71%	100%	83%
	DrDoS_MSSQL	4,524,498	53.85%	50%	100%	67%
	DrDoS_NetBIOS	4,094,986	14.29%	14%	100%	25%
	DrDoS_NTP	1,217,007	100%	100%	100%	100%
	DrDoS_SNMP	5,161,377	66.67%	74%	82%	78%
	DrDoS_SSDP	2,611,374	75.00%	100.00%	75.00%	86%
	DrDoS_UDP	3,136,802	33.33%	80%	36%	50%
	Syn	1,582,681	66.67%	66.67%	100%	80%
2nd Day	LDAP	2,113,234	85.71%	83%	100%	91%
	MSSQL	5,775,786	68%	62%	89%	77%
	NetBIOS	3,455,899	100%	100%	100%	100%
	Portmap	191,694	100%	100%	100%	100%
	Syn	4,320,541	100%	100%	100%	100%
	UDP	3,782,206	100%	100%	100%	100%
	UDPLag	725,165	100%	100%	100%	100%
Mixed files	DrDoS_LDAP, UDP	2,906,707	89.47%	100%	89%	94%
	Syn, UDP	5,364,887	50.00%	100%	50%	67%
	DrDoS_MSSQL,MSSQL	10,300,284	50.00%	56%	83%	67%
Average			73.32%	79.83%	89.3%	77.0%

TABLE III. ZF-DDOS DETAILED EVALUATION RESULTS

Day	File	Number of records	Accuracy		Precision		Recall		F1-Score	
			TCP	*UDP*	*TCP*	*UDP*	*TCP*	*UDP*	*TCP*	*UDP*
1st Day	DrDoS_DNS	4,320,541	91.5%	91.5%	88.5%	88.5%	94.4%	94.4%	91.4%	91.4%
	DrDoS_LDAP	2,181,542	91.5%	91.5%	88.5%	88.5%	94.4%	94.4%	91.4%	91.4%
	DrDoS_MSSQL	4,524,498	91.5%	91.5%	88.5%	88.5%	94.4%	94.4%	91.4%	91.4%
	DrDoS_NetBIOS	4,094,986	91.5%	91.5%	88.5%	88.5%	94.4%	94.4%	91.4%	91.4%
	DrDoS_NTP	1,217,007	91.5%	91.5%	88.5%	88.5%	94.4%	94.4%	91.4%	91.4%
	DrDoS_SNMP	5,161,377	91.5%	91.5%	88.5%	88.5%	94.4%	94.4%	91.4%	91.4%
	DrDoS_SSDP	2,611,374	91.5%	91.5%	88.5%	88.5%	94.4%	94.4%	91.4%	91.4%
	DrDoS_UDP	3,136,802	91.5%	91.5%	88.5%	88.5%	94.4%	94.4%	91.4%	91.4%
	Syn	1,582,681	91.5%	91.5%	88.5%	88.5%	94.4%	94.4%	91.4%	91.4%
2nd Day	LDAP	2,113,234	91.5%	91.5%	88.5%	88.5%	94.4%	94.4%	91.4%	91.4%
	MSSQL	5,775,786	91.5%	91.5%	88.5%	88.5%	94.4%	94.4%	91.4%	91.4%
	NetBIOS	3,455,899	91.5%	91.5%	88.5%	88.5%	94.4%	94.4%	91.4%	91.4%
	Portmap	191,694	91.5%	91.5%	88.5%	88.5%	94.4%	94.4%	91.4%	91.4%
	Syn	4,320,541	91.5%	91.5%	88.5%	88.5%	94.4%	94.4%	91.4%	91.4%
	UDP	3,782,206	91.5%	91.5%	88.5%	88.5%	94.4%	94.4%	91.4%	91.4%
	UDPLag	725,165	91.5%	91.5%	88.5%	88.5%	94.4%	94.4%	91.4%	91.4%
Mixed files	DrDoS_LDAP,UDPLag	2,906,707	91.5%	91.5%	88.5%	88.5%	94.4%	94.4%	91.4%	91.4%
	Syn,UDP	5,364,887	91.5%	91.5%	88.5%	88.5%	94.4%	94.4%	91.4%	91.4%
	DrDoS_MSSQL,MSSQL	10,300,284	91.5%	91.5%	88.5%	88.5%	94.4%	94.4%	91.4%	91.4%
Average			91.5%	91.5%	92.6%	92.6%	90.9%	90.9%	91.8%	91.8%

TABLE IV. AVERAGE DETAILED EVALUATION RESULTS

Model	Accuracy	Precision		Recall		F1-Score	
		TCP	*UDP*	*TCP*	*UDP*	*TCP*	*UDP*
Z-Score	83.45%	87.7%	0%	83.3%	0%	85.5%	0%
Fast-Entropy	73.32%	79.83%	79.83%	89.39%	89.39%	77.0%	77.0%
ZD-DDoS	**91.5%**	**92.6%**	**92.6%**	**90.9%**	**90.9%**	**91.8%**	**91.8%**

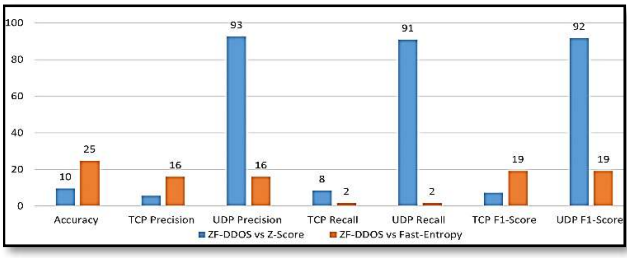


Fig.3. Enhancement Ratios of the Proposed ZF-DDoS Hybrid Approach

The targeted dataset CIC-DDOS-2019 presents the real case of DDoS attacks on both TCP and UDP traffic, which cannot be efficiently detected by one of the statistical approaches(Z-Score or Fast-Entropy) but can be efficiently detected with our ZF-DDOS approach model.

V. CONCLUSION AND FUTURE WORK

This paper proposed an enhanced ZF-DDOS detection approach that integrated two statistical measures (Z-Score, Fast-Entropy). In addition, new calculated feature has been added to the proposed ZF-DDOS detection. Using hybrid combination of statistical measures together helps improve the certainty and assurance of detection of DDoS attack in the network. Moreover, this guarantee additional characteristics to the proposed approach as follows: Complementary Use; Fast-Entropy and Z-Score can be used together to complement each other's strengths and weaknesses. For example, Fast-Entropy may be suitable for detecting sudden spikes in flow counts, while Z-Score may

Authorized licensed use limited to: UNIVERSIDADE DE BRASILIA. Downloaded on October 22,2026 at 00:29:27 UTC from IEEE Xplore. Restrictions apply.

excel in identifying deviations in other attributes. Continuous Monitoring; Both methods are suitable for continuous monitoring of network traffic, allowing for the timely detection of anomalies.

Although statistical methods are very primitive methods, however, it is considered as a light weight technique with fast response, low processing speed and computation overheads compared to ML/DL based techniques. From the experimental results, it is clear that ZF-DDOS achieved an accuracy enhancement over the Fast-Entropy model with 25% and about 10% over the Z-Score model. It's important to note that the effectiveness of these approach can depend on the specific characteristics of the network, the nature of the DDoS attacks, and the chosen parameters. Experimentation and testing with real-world data are essential to determine the most suitable approach for a particular use case.

VI. References

[1] M. Mittal, K. Kumar, and S. Behal, "Deep learning approaches for detecting DDoS attacks: a systematic review, " Soft Comput, vol. 27, pp.13039–13075,2023..https://doi.org/10.1007/s00500-021-06608-1.

[2] S. Malliga, P. S. Nandhini, and S. V. Kogilavani, "A Comprehensive Review of Deep Learning Techniques for the Detection of (Distributed) Denial of Service Attacks, " Information Technology andControl,vol.51,no.1,pp.180-215,2022. https://doi.org/10.5755/j01.itc.51.1.29595.

[3] NetScout, "DDoS Vector Discovery and Attack Enablers," ThreatIntelligence .https://www.netscout.com/threatreport/ddos-vector-discovery-and-attack-enablers.

[4] Kaspersky, "DDoS Attacks in Q3 2022, "Securelist. ". https://securelist.com/ddos-report-q3-2022/107860/.

[5] A. Khraisat et al., "Survey of intrusion detection systems: Techniques, datasets and challenges, " Cybersecur, vol. 2, no. 1, p. 20, 2019.

[6] A. B. de Neira, B. Kantarci, and M. Nogueira, "Distributed denial of service attack prediction: Challenges, open issues and opportunities, " Computer Networks, vol. 222, pp. 109553, 2023. https://doi.org/10.1016/j.comnet.2022.109553.

[7] R. F. Fouladi, O. Ermiş, and E. Anarim, "A DDoS attack detection and defense scheme using time-series analysis for SDN, " Journal of Information Security and Applications, vol. 54, 2020, pp. 102587. https://doi.org/10.1016/j.jisa.2020.102587.

[8] M. Nooribakhsh and M. Mollamotalebi, "A review on statistical approaches for anomaly detection in DDoS attacks, " Information Security Journal: A Global Perspective, vol. 29, no. 3, pp. 118-133, 2020. https://doi.org/10.1080/19393555.2020.1717019.

[9] T. E. Ali, Y.-W. Chong, and S. Manickam, "Machine Learning Techniques to Detect a DDoS Attack in SDN: A Systematic Review,"Appl.Sci.,vol.13,pp.3183,2023. https://doi.org/10.3390/app1305318.

[10] I. O. Lopes, D. Zou, F. A. Ruambo, S. Akbar, and B. Yuan, "Towards Effective Detection of Recent DDoS Attacks: A Deep Learning Approach, " Security and Communication Networks, vol. 2021, pp.1-14, 2021. https://doi.org/10.1155/2021/5710028.

[11] M. Najafimehr, S. Zarifzadeh, and S. Mostafavi, "DDoS attacks and machine-learning-based detection methods: A survey and taxonomy," EngineeringReports, vol.5, no.12, e12697, 2023. https://doi.org/10.1002/eng2.12697.

[12] J. David and C. Thomas, "DDoS Attack Detection Using Fast-Entropy Approach on Flow-Based Network Traffic, " Procedia Computer Science, vol.50, pp.30-36, 2015. ISSN:1877-0509. https://doi.org/10.1016/j.procs.2015.04.007.

[13] P. Hick, E. Aben, K. Claffy, and J. Polterock, "The CAIDA 'DDoS Attack 2007' Dataset. ".

[14] H. Majed, H. Noura, O. Salman, M. Malli, and A. Chehab, "Efficient and Secure Statistical DDoS Detection Scheme, " in the 6th International Conference on Information Systems Security and Privacy,2020,pp.153-161. https://doi.org/10.5220/0009873801530161.

[15] M. Aladaileh, M. Anbar, I. H. Hasbullah, Y. K. Sanjalawe, and Y. Chong, "Entropy-based approach to detect DDoS attacks on software-defined networking controller, " Computers, Materials & Continua, vol. 69, no. 1, pp. 373–391, 2021.

[16] B. Bouyeddou, B. Kadri, F. Harrou, and Y. Sun, "DDoS-attacks detection using an efficient measurement-based statistical mechanism, " Engineering Science and Technology, an International Journal, vol.23,no.4,pp.870-878,2020.ISSN:2215-0986. https://doi.org/10.1016/j.jestch.2020.05.002.

[17] DARPA, "DARPA 1999 Intrusion Detection Evaluation Data, " [Online].Available:www.ll.mit.edu/ideval/data/1999data.html. Accessed: December 27, 2023 [DARPA99].

[18] MAWI, "MAWI (Measurement and Analysis on the WIDE Internet) Datasets, Available: http://www.fukudalab.org/mawilab/data.html Accessed: December 27, 2023 [MAWI].

[19] O. E. Elejla, B. Belaton, M. Anbar, and A. Alnajjar, "A reference dataset for ICMPv6 flooding attacks, " Journal of Engineering and Applied Sciences, vol. 11, no. 3, pp. 476–481, 2016.

[20] K. Kalkan et al., "JESS: Joint Entropy-Based DDoS Defense Scheme in SDN, " IEEE Journal on Selected Areas in Communications, vol. 36, pp. 2358-2372, 2018.

[21] R. Hajtmanek, M. Kontšek, J. Smieško, and J. Uramová, "One-Parameter Statistical Methods to Recognize DDoS Attacks," Symmetry, vol. 14, pp. 2388, 2022. https://doi.org/10.3390/sym14112388.

[22] D. Erhan, S. Özdel, and E. Anarim, "DDoS Detection Using Statistical Modelling, " in International Conference on Digital Image & Signal Processing (DISP'19), 2019.

[23] J. Zhang, Q. Liang, R. Jiang, and X. Li, "A Feature Analysis Based Identifying Scheme Using GBDT for DDoS with Multiple Attack Vectors, "Applied Sciences,vol.9,no.21,pp.4633,2019. https://doi.org/10.3390/app9214633.

[24] X. Gao, C. Shan, C. Hu, Z. Niu, and Z. Liu, "An Adaptive Ensemble Machine Learning Model for Intrusion Detection, " IEEE Access, vol.7, pp. 82512-82521, 2019. https://doi.org/10.1109/ACCESS.2019.2923404.

[25] D. Akgun, S. Hizal, U. Cavusoglu, "A New DDoS Attacks Intrusion Detection Model Based on Deep Learning for Cybersecurity, " Computers & Security, vol. 118, pp. 102748, 2022. https://doi.org/10.1016/j.cose.2022.102748.

[26] J. Halladay et al., "Detection and Characterization of DDoS Attacks Using Time-Based Features, " in IEEE Access, vol. 10, pp. 49794-49807, 2022. https://doi.org/10.1109/ACCESS.2022.3173319.

[27] S. Batool et al., "Lightweight Statistical Approach towards TCP SYN Flood DDoS Attack Detection and Mitigation in SDN Environment, " Security and Communication Networks, vol. 2022, pp. 1-14, 2022, https://doi.org/10.1155/2022/2593672.

[28] Ö. Cepheli, S. Büyükçorak, and G. K. Kurt, "Hybrid Intrusion Detection System for DDoS Attacks, "Journal of Electrical and Computer Engineering, vol. 2016, Article ID 1075648, pp. 1-8, 2016. https://doi.org/10.1155/2016/1075648.

[29] M. A. Aladaileh et al., "Effectiveness of an Entropy-Based Approach for Detecting Low- and High-Rate DDoS Attacks against the SDN Controller: Experimental Analysis, " Applied Sciences, vol. 13, no. 2, pp. 775, 2023, https://doi.org/10.3390/app13020775.

[30] I. Sharafaldin, A. Habibi Lashkari, S. Hakak, and A. A. Ghorbani, "Developing Realistic Distributed Denial of Service (DDoS) Attack Dataset and Taxonomy, " in IEEE 53rd International Carnahan Conference on Security Technology, Chennai, India, 2019.

Authorized licensed use limited to: UNIVERSIDADE DE BRASILIA. Downloaded on October 22,2026 at 00:29:27 UTC from IEEE Xplore. Restrictions apply.